

- 5 Fig. 5.1 shows an electron passing an earthed vertical plate P_1 with negligible speed and accelerates horizontally towards vertical plate P_2 which is at a potential 200 V.

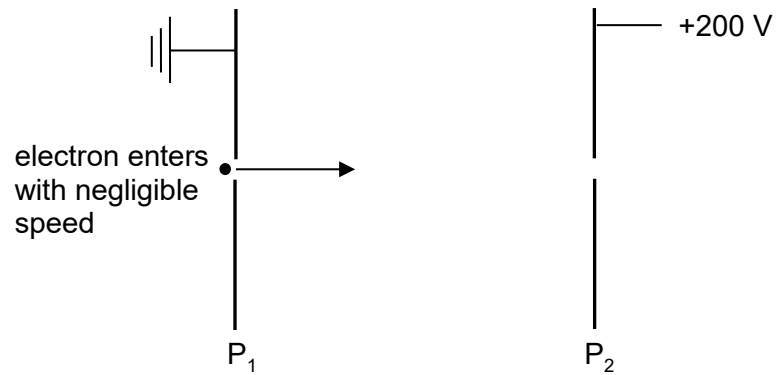


Fig. 5.1

- (a) Show that the speed of the electron when it reaches plate P_2 is $8.38 \times 10^6 \text{ m s}^{-1}$.

[2]

- (b) After emerging from plate P_2 , the electron enters a region of uniform magnetic field of flux density B , directed into the plane of the paper as shown in Fig. 5.2.

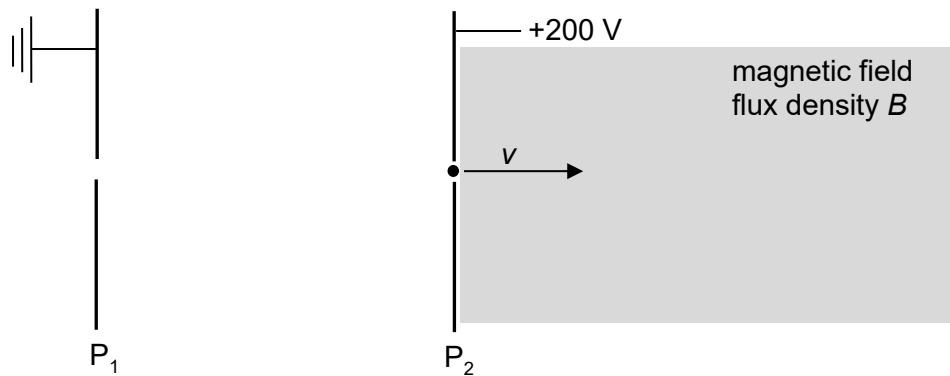


Fig. 5.2

- (i) On Fig. 5.2, draw a solid line to represent a possible path of the electron. [1]

- (ii) The electron moves in a path with a radius of curvature of 2.8 cm.

Calculate the magnitude of the magnetic flux density B .

$$B = \dots\dots\dots \text{ T [2]}$$

- (c) The direction of the magnetic field is now changed such that it is at an angle of 30° below the horizontal, as shown in Fig. 5.3.

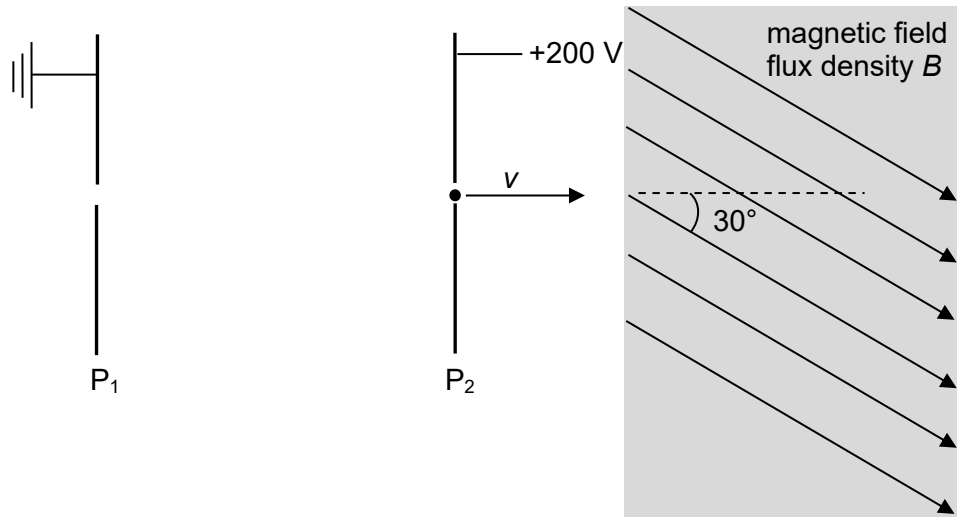


Fig. 5.3

- (i) By considering the components of the velocity parallel to the magnetic field and at right-angle to the magnetic field, describe and explain qualitatively the motion of the electron in the field.

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..... [3]

- (ii) Calculate the radius of curvature of this new path.

radius = m [2]

6 (a) Fig. 6.1 shows the energy levels of sodium gas